



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

AIOT

12-05-2024

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY

PAPER-1 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C,D	A,C	A,B,C	A,D	A,C	B,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	C	D	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.25	800.00	0.50	1.00	0.50	2.00	0.50	0.20

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,C	A,C,D	B,D	D	B,C	A,B,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	D	D	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	4.00	5.00	3.00	14.00	0.75	44.00	1170.00	1.50

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C	A,B,C	A,B,C,D	A,D	A,C,D	A,B,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	B	A	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	1.60	2.25	2.25	16.00	14.40	9.66 or 9.67	30.00	2.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,B,C,D)

$$I = \frac{V_o}{R}$$

$$q = C_o V_o - \frac{V_o t}{R}$$

$$C = \frac{q}{V_o} = C_o \left(1 - \frac{t}{RC_o} \right)$$

$$\text{Heat} = I^2 R \times \tau$$

2. Ans (A,C)

$$2mg \sin \alpha = mg \sin \beta$$

$$\Rightarrow 2 \sin \alpha = \sin \beta$$

When length of string is maximum

$$\text{then } \sin \beta = 1 \text{ and } \sin \alpha = \frac{1}{2}$$

$$\Rightarrow \alpha + \beta = \frac{2\pi}{3}$$

$$\therefore \ell_{\max} = R(\alpha + \beta) = \frac{2\pi R}{3}$$

3. Ans (A,B,C)

Conceptual.

4. Ans (A,D)

Point A and C are on the same line passing

$$\text{through origin} \Rightarrow \frac{P_A}{V_A} = \frac{P_C}{V_C} \quad \dots(1)$$

$$\text{Also } T_A = 200 \text{ K} = \frac{P_A V_A}{nR} \text{ and Also } T_C = 1800$$

$$K = \frac{P_C V_C}{nR} \Rightarrow \frac{P_A V_A}{P_C V_C} = \frac{1}{9} \quad \dots(2)$$

$$\text{From (1) and (2)} \quad \frac{V_A}{V_C} = \frac{1}{3}$$

5. Ans (A,C)

From Mosley law,

$$\sqrt{\frac{1}{\lambda}} \propto (z-1) \text{ for } K_{\alpha} \text{ line}$$

$$\therefore \frac{\lambda_z}{\lambda_1} = \left(\frac{z_{\text{imp1}} - 1}{z_{\text{target}} - 1} \right)^2 \quad \dots\dots (i)$$

$$\text{for 2nd impurity} \quad \frac{\lambda_z}{\lambda_2} = \left(\frac{z_{\text{imp2}} - 1}{z_{\text{target}} - 1} \right)^2 \quad \dots\dots (ii)$$

6. Ans (B,D)

For (B) : Force of buoyancy increases.

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (B)

$$F = mg \sin \theta + \mu_k mg \cos \theta \text{ and } p = \vec{F} \cdot \vec{V}$$

8. Ans (C)

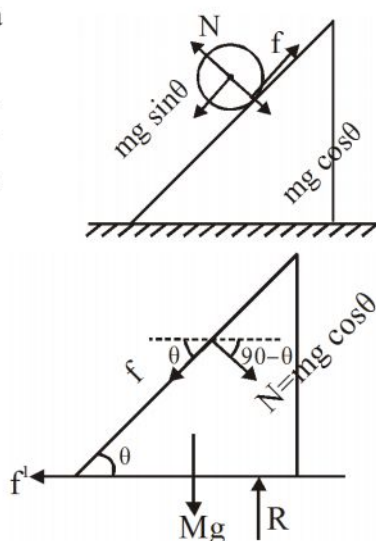
$$mg \sin \theta - f = ma$$

$$fR = \frac{MR^2}{2} \cdot \frac{a}{R}$$

$$mg \sin \theta = \frac{3}{2} ma$$

$$a = \frac{2g \sin \theta}{3}$$

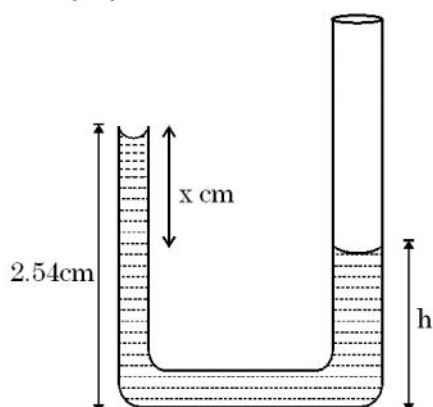
$$f = \frac{mg \sin \theta}{3}$$



9. Ans (D)

Distant object subtend small angle at eye, so they appear moving along observer.

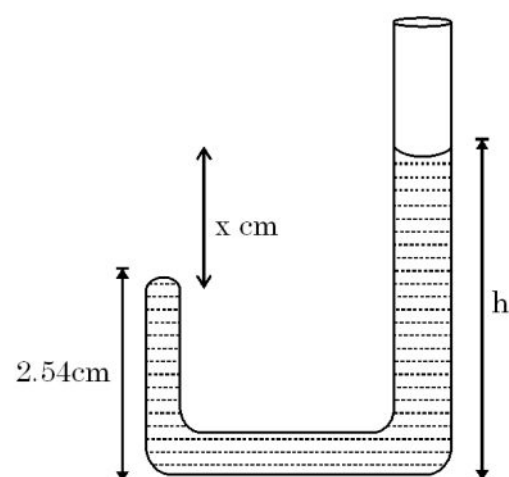
10. Ans (C)



$$x = \frac{2T}{\rho g} \left[\frac{2}{d_1} - \frac{2}{d_2} \right] = 1.44 \text{ cm}, \text{ so } h = 1.1 \text{ cm}$$

When $h > 1.1 \text{ cm}$, radius of meniscus in left limb first increases and then becomes flat for some value of h . For further increase in height, the meniscus in the left limb becomes convex upward and in limiting case the radius of the curvature of the meniscus becomes equal to radius of the capillary.

Under this condition



$$P_0 + \frac{4T}{d_1} = P_0 - \frac{4T}{d_2} + x\rho g$$

$$x = 4.32 \text{ cm}, h = 4.32 + 2.54 = 6.86 \text{ cm}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (0.25)**

$$I_x = I \frac{x^2}{r^2}, R_x = \frac{i\ell}{\pi x^2}$$

$$\therefore P = \frac{I^2 x^2 i \ell}{\pi r^4}$$

$$\text{Also } P = -2\pi x K \frac{dT}{dx}$$

$$\Rightarrow \frac{dT}{dx} = -\frac{I^2 \rho}{2\pi^2 r^4 K} x$$

$$\therefore T = T_0 + \frac{I^2 \rho}{4\pi^2 r^2 K}$$

2. **Ans (800.00)**

$$Q = 800 \times 20 = C \Delta T$$

$$\therefore C = \frac{16000}{20} = 800 \text{ J/K}$$

3. **Ans (0.50)**

$$T = \frac{2m(M-m)g}{M}$$

$$\frac{dT}{dm} = 0 \Rightarrow m = \frac{M}{2}$$

4. **Ans (1.00)**

$$v^2 = I \Rightarrow (v_0 - Ir)^2 = I$$

5. **Ans (0.50)**

$$Z_{RL} = \sqrt{R^2 + \omega^2 L^2}$$

$$Z_{RLC} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2}$$

$$\cos \phi_{RL} = \frac{R}{\sqrt{R^2 + \omega^2 L^2}}$$

$$\cos \phi_{RLC} = \frac{R}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2}}$$

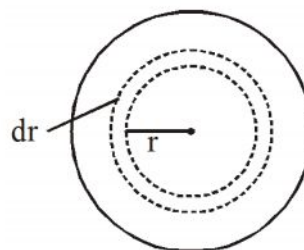
According to question

$$(\omega L)^2 = \left(\omega L - \frac{1}{C\omega} \right)^2 \Rightarrow C = \frac{1}{2\omega^2 L}$$

6. **Ans (2.00)**

$$dm = \frac{M}{\frac{4}{3}\pi R_0^3} \cdot 4\pi r^2 dr$$

$$dm = \frac{3M}{R_0^3} \cdot r^2 dr, V = \frac{r}{R_0} V_0$$



$$K.E. = \frac{1}{2} \int dm \left(\frac{rV_0}{R_0} \right)^2$$

$$= \int \frac{1}{2} \frac{3M}{R_0^3} r^2 dr \cdot \frac{r^2 V_0^2}{R_0^2}$$

$$= \int \frac{3MV_0^2}{2R_0^5} r^4 dr = \frac{3MV_0^2}{10}, \quad \frac{3MV_0^2}{10} = \Delta U$$

$$= \frac{3}{5} GM^2 \left[\frac{1}{R_0} - \frac{1}{R} \right]$$

$$R = \frac{2GMR_0}{2GM - R_0 V_0^2}$$

7. **Ans (0.50)**

$$\phi = (\vec{E} \cdot \hat{n})A = \left[\frac{(8 \times 2) + (6 \times 3) + (10 \times 6)}{\sqrt{2^2 + 3^2 + 6^2}} \right] \left(\frac{7}{47} \right)$$

8. **Ans (0.20)**

At $x = 31.5 \text{ mm}$, both $3600 \text{ \AA}$ and $6000 \text{ \AA}$

produce minima. If the slit separation is d , then,

$$\frac{xd}{D\lambda_1} = n_1 + \frac{1}{2}, \quad \frac{xd}{D\lambda_2} = n_2 + \frac{1}{2}$$

$$\text{or } \frac{xd}{D} \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right) = n_1 - n_2 = 7$$

Substituting the values, $D = 1\text{m}$, $\lambda_1 = 3600\text{\AA}$, λ_2

$= 6000 \text{ \AA}$

we get $d = 0.2 \text{ mm}$.

PART-2 : CHEMISTRY

SECTION-I (ii)

9. Ans (D)

(P) $P = \text{low}, T < T_B$

$\therefore Z < 1$ or $PV \neq nRT$

Gas is more compressible with respect to an ideal gas.

$\therefore T > T_c$, so gas cannot be liquefied at any pressure at given temperature.

$$(V_m)_{\text{real}} < (V_m)_{\text{ideal}}$$

(Q) $P = \text{very high}, T = T_B$

$\therefore Z > 1$ or $PV \neq nRT$

$\therefore T > T_c$, so gas cannot be liquefied at any pressure at given temperature.

$$(V_m)_{\text{real}} > (V_m)_{\text{ideal}}$$

(R) $P = \text{low}, T = T_B$

$\therefore Z > 1$ or $PV = nRT$

$\therefore T > T_c$, so gas cannot be liquefied at any pressure at given temperature.

10. Ans (D)

(P) $K_{sp} = 4s^3 \Rightarrow s = 10^{-4} \text{ M} \Rightarrow [\text{OH}^-] = 2 \times 10^{-4}$

$$\therefore \text{pH} = 10 + \log 2$$

(Q) Basic buffer is formed. $\therefore \text{pH} = 14 - 4 - \log$

$$\left[\frac{10}{20} \right] = 10 + \log 2$$

(R) A salt of weak acid and weak base is formed in

which $k_a = k_b$

Hence, $\text{pH} = 7$ or

$$\text{pH} = \frac{1}{2} \text{p}K_w + \frac{1}{2} \text{p}K_a - \frac{1}{2} \text{p}K_b$$

(S) $\text{H}_3\text{PO}_4 + \text{H}_2\text{PO}_4^-; K_{a1}$



p^H of $0.01 \text{ M Na}_2\text{HPO}_4 =$

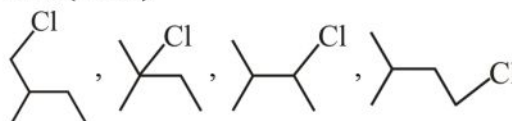
$$\frac{\text{p}K_{a2} + \text{p}K_{a3}}{2} = \frac{8.102 + 12.5}{2}$$

$$= \frac{20.602}{2} = 10.301$$

PART-2 : CHEMISTRY

SECTION-II

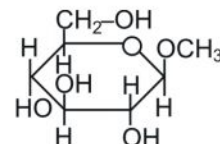
1. Ans (4.00)



2. Ans (5.00)

Reducing sugars are those sugars in which anomeric carbon atom has $-\text{OH}$

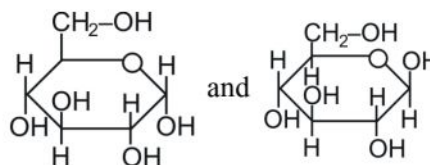
group (Hemiacetal form)



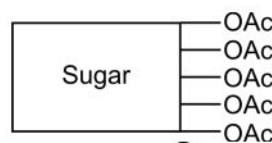
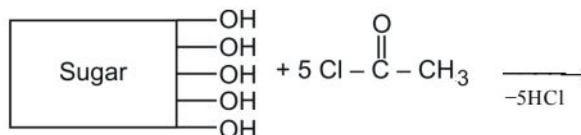
is non reducing sugar (rest all are reducing)

$$x = 7$$

$\rightarrow$ Only these two sugars will be form pentactate because five $-\text{OH}$ groups are present



$$y = 2$$

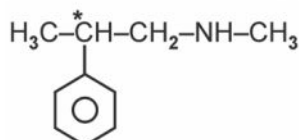
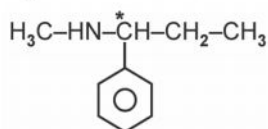
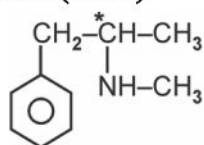


Addition of one $-\text{C}(=\text{O})-\text{CH}_3$ group will change in mol. wt. by 42 units

$$\therefore \text{Number of the } \text{CH}_3-\text{C}(=\text{O})- \text{ groups added} = \frac{210}{42} = 5$$

$$x - y = 7 - 2 = 5$$

3. Ans (3.00)



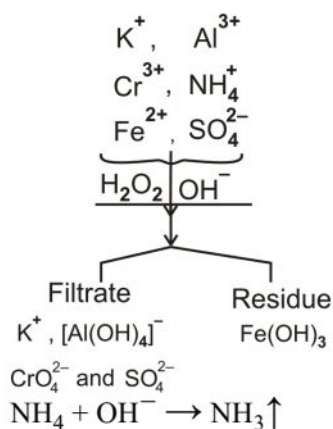
4. Ans (14.00)

Potash Alum : $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$

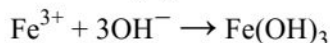
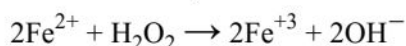
Chrome Alum : $\text{K}_2\text{SO}_4 \cdot \text{Cr}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$

Mohr's salt : $(\text{NH}_4)_2 \cdot \text{FeSO}_4 \cdot 6\text{H}_2\text{O}$

In solution :-



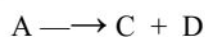
will be out of system.



Mass of Fe : 56 = a

$$b = 4, a/b = 56/4 = 14.00$$

6. Ans (44.00)



$$t = 0 \quad 8 \quad 0 \quad 0$$

$$t = 100 \text{ min.} \quad 2 \quad 6 \quad 6$$

$$n_{\text{total}} = 2 + 6 + 6 + 4 + 2 = 20 \text{ mole.}$$

$$P_T = P_A^\circ X_A + P_B^\circ X_B + P_C^\circ X_C + P_D^\circ X_D$$

$$= 90 \times \frac{2}{20} + 40 \times \frac{4}{20} + 50 \times \frac{6}{20} + 40 \times \frac{6}{20}$$

$$= 9 + 8 + 15 + 12 = \boxed{44 \text{ torr}}$$

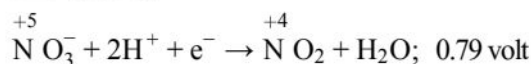
7. Ans (1170.00)

$$\Delta H = \Delta U + V \Delta P$$

$$= 350 + 24.6 \left(\frac{4}{3} - 1 \right) \times 100$$

$$= 1170$$

8. Ans (1.50)



$$E = 0.79 - \frac{0.0591}{1} \log \frac{1}{[\text{H}^+]^2} = 0.731 - \frac{0.0591}{6} \log \frac{1}{[\text{H}^+]^8}$$

$$0.79 + 2 \times \frac{0.0591}{1} \log [\text{H}^+]$$

$$= 0.731 + \frac{0.0591}{6} \times 8 \times \log [\text{H}^+]$$

$$0.059 = 0.059 \left(2 - \frac{8}{6} \right) \text{ pH}$$

$$\text{pH} = \frac{6}{4} = 1.5$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,B,C)

$$f(x) = ax^2 - bx + c$$

$$\text{S.R.} = \frac{b}{a} < 2 \Rightarrow b < 2a \quad \dots(1)$$

$$\text{P.R.} = \frac{c}{a} < 1 \Rightarrow c < a \quad \dots(2)$$

$$\text{and } D = b^2 - 4ac > 0 \Rightarrow$$

$$b^2 > 4ac \quad \dots(3)$$

To find minimum value of abc

$$f(0) = c > 0; C_{\min.} = 1$$

$$\therefore (3) \Rightarrow b^2 > 4a$$

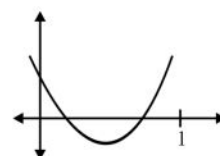
$$(1) \Rightarrow b > 2a$$

$$\Rightarrow b = 5; a = 5; c = 1$$

$$\text{Also, } f(1) > 0 \Rightarrow a - b + c > 0; a + 1 > b$$

$$\lambda \leq \log_5 25$$

$$\lambda \leq 2$$



2. Ans (A,B,C)

$$P(A \cap B) = P(A) P(B)$$

$$P(A \cap (B \cap C)) = P(A)P(B \cap C)$$

$$P(A \cap (B \cup C)) = P(A) P(B \cup C)$$

$$\Rightarrow P(A \cap B) + P(A \cap C) - P(A \cap B \cap C)$$

$$= P(A)[P(B) + P(C) - P(B \cap C)]$$

$$\Rightarrow P(A)P(B) + P(A \cap C) - P(A)P(B \cap C)$$

$$= P(A)P(B) + P(A)P(C) - P(A)P(B \cap C)$$

$$\Rightarrow P(A \cap C) = P(A)P(C)$$

$\Rightarrow$ A and C are independent events.

$$\Rightarrow P\left(\frac{A}{C}\right) = P(A) = \frac{1}{2}$$

$$P\left(\frac{\overline{B \cup C}}{A}\right) = P\left(\frac{\overline{B \cap C}}{A}\right) = P\left(\frac{\overline{B \cap C}}{A}\right)$$

$$= 1 - P(B)P(C) = 1 - \frac{1}{3} \cdot \frac{1}{4} = \frac{11}{12}$$

$$P\left(\frac{\overline{A}}{\overline{B \cap C}}\right) = P(\overline{A}) = \frac{1}{2}$$

3. Ans (A,B,C,D)

$\because$ P(3,4) is foot of perpendicular from S (0, 0)

on $3x + 4y - 25 = 0$

which is also on ellipse

$\therefore$ P is vertex of ellipse.

$\therefore$ Distance between focus and directrix = $a - ae$
= 5

$$\Rightarrow a - \frac{a}{2} = 5 \Rightarrow a = 10$$

$$\because b^2 = a^2(1 - e^2) = 100\left(1 - \frac{1}{4}\right) = 75$$

Clearly AB will be latus rectum of ellipse.

$$\therefore AB = \text{length of LR} = \frac{2b^2}{a} = 15$$

$$\therefore \text{Focal length} = 2ae = 10$$

Mid-point of vertex and centre will focus,

$$\therefore \text{centre} = (-3, -4)$$

4. Ans (A,D)

$$(A) T_n^3 + T_n + 1 = T_{n+1} - T_n$$

$$\sum_{n=1}^{100} (T_n^3 + T_n + 1) = T_{101} - T_1 + 100 = T_{101} + 99$$

$$(C) T_n^2 + 2 = \frac{T_{n+1}}{T_n}$$

$$\prod_{n=1}^{100} (T_n^2 + 2) = \prod_{n=1}^{100} \frac{T_{n+1}}{T_n} = \frac{T_{101}}{T_1} = T_{101}$$

$$(D) \prod_{n=1}^{100} \left(\frac{T_{n+1}}{T_n} - T_n^2 \right) = 2^{100}$$

5. Ans (A,C,D)

$$f(xy) = e^{xy-x} \cdot f(x) + e^{xy-y} f(y)$$

Differentiating w.r.t. x

$$yf'(xy) = e^{xy-x} f'(x) + f(x)e^{xy-x} (y - 1) +$$

$$f(y)e^{xy-y} \cdot y$$

Put $x = 1$

$$yf'(y) = e^{y-1} f'(1) + f(1)e^{y-1} (y - 1) + yf(y)$$

Now put $x = y = 1$; $f(1) = 0$

$$\therefore yf'(y) = e^y + yf(y) \Rightarrow x(f'(x) - f(x)) = e^x$$

$$e^{-x}(f'(x) - f(x)) = \frac{1}{x}$$

Integrate both sides

$$e^{-x}f(x) = \ln x + C$$

$$f(1) = 0 \quad \therefore C = 0$$

$$\Rightarrow f(x) = e^x \ln x. \text{ Now verify.}$$

6. Ans (A,B,C,D)

Three planes meet at two points it means they

have infinitely many solutions, so

$$\begin{vmatrix} 2 & 1 & 1 \\ 1 & -1 & 1 \\ \alpha & -1 & 3 \end{vmatrix} = 0$$

$$\Rightarrow 2(-3+1) - 1(3+1) + \alpha(1+1) = 0$$

$$\Rightarrow \alpha = 4$$

$$P_1 : 2x + y + z = 1$$

$$P_2 : x - y + z = 2$$

$$P_3 : 4x - y + 3z = 5$$

$$P \text{ on XOY plane} \equiv (1, -1, 0)$$

(which can be obtained by putting $z = 0$ in any two of the given planes.)

$$Q \text{ on YOZ plane} \equiv \left(0, \frac{-1}{2}, \frac{3}{2}\right)$$

(which can be obtained by putting $x = 0$ in any two of the given planes.)

$\therefore$ Straight line perpendicular to plane

P_3 passing through P is :

$$\frac{x-1}{4} = \frac{y+1}{-1} = \frac{z}{3}$$

$$\vec{PQ} = \hat{i} - \frac{1}{2}\hat{j} - \frac{3}{2}\hat{k}$$

$$\text{Projection of } \vec{PQ} \text{ on x-axis} \Rightarrow \left| \frac{\vec{OP} \cdot \hat{i}}{|\hat{i}|} \right| = 1$$

$$\text{Centroid of } \triangle OPQ \text{ is } \left(\frac{1}{3}, \frac{-1}{2}, \frac{1}{2}\right)$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (A)

(I) 20, b, c are in H.P.

$$\Rightarrow \frac{1}{20} + \frac{1}{c} = \frac{2}{b}$$

This can be written as $(40 - b)(c + 20) = 800$ we must have $20 < b < 40$ or $0 < 40 - b < 20$,

consider factorisation of 800 in which one term is less than 20

$$(40 - b)(c + 20) = 800 = 1 \times 800 = 2 \times 400 = 4 \times 200 = 5 \times 160 = 8 \times 100 = 10 \times 80 = 16 \times 50$$

Thus pair (b, c) = (39, 780), (38, 380), (36, 180), (35, 140), (32, 80), (30, 60), (24, 30) but b divides c

$\Rightarrow$ Only 5 triples satisfy required requirement

(II) Solution of $m + n + p = 10$, are ${}^{3+10-1}C_{3-1}$

(III) $ab + b - a + 1 = 0$

$$\Rightarrow (a + 1)(b - 1) = -2$$

$$\Rightarrow (a + 1, b - 1) \in \{(1, -2), (-1, 2), (2, -1), (-2, 1)\}$$

$$\Rightarrow (a, b) \in \{(0, -1), (-2, 3), (1, 0), (-3, 2)\}$$

(IV) Let $z = x + iy$ then $(x + iy)^{20} = x - iy$

$$\Rightarrow |z|^{20} = |z| \Rightarrow |z|(|z|^{19} - 1) = 0$$

$$\Rightarrow |z| = 1 \text{ or } 0$$

$$\Rightarrow |z| = 0$$

$\therefore$ Ordered pair (0, 0)

Let $z \neq 0$ then $(x + iy)^{20}(x + iy) = x^2 + y^2$

$$\Rightarrow z^{21} = |z|^2 = 1 \Rightarrow z^{21} - 1 = 0, \text{ which are 21 roots}$$

8. Ans (B)

$$(I) f(x) = x^4 + ax^3 + bx^2 + cx + d$$

$\alpha, \beta, \gamma, \delta$ are real roots of equation $f(x) = 0$

$$\Rightarrow f(x) = (x - \alpha)(x - \beta)(x - \gamma)(x - \delta)$$

$$\Rightarrow |f(i)| = 1 \Rightarrow (1 + \alpha^2)(1 + \beta^2)(1 + \gamma^2)(1 + \delta^2)$$

$$= 1 \text{ this is possible if } \alpha = \beta = \gamma = \delta = 0$$

$$\Rightarrow a = b = c = d = 0$$

$$(II) x^5 - 1 = 0$$

$$\Rightarrow (x - 1)(x^4 + x^3 + x^2 + x + 1) = 0$$

$$\begin{aligned} \Rightarrow \frac{x}{1+x^2} + \frac{x^2}{1+x^4} + \frac{x^3}{1+x} + \frac{x^4}{1+x^3} \\ = \frac{x}{1+x^2} + \frac{x^2}{1+x^4} + \frac{x^3}{x^5+x} + \frac{x^4}{x^5+x^3} \\ = \frac{x}{1+x^2} + \frac{x^2}{1+x^4} + \frac{x^2}{1+x^4} + \frac{x}{1+x^2} \\ = 2 \left(\frac{x}{1+x^2} + \frac{x^2}{1+x^4} \right) = \frac{2(x+x^5+x^2+x^4)}{1+x^2+x^4+x^6} = 2 \end{aligned}$$

$$(III) \left[\cos|x| + \frac{1}{2} \right] > 0 \Rightarrow \cos|x| + \frac{1}{2} \geq 1 \text{ and}$$

$$-\frac{1}{2} \leq \cos|x| + \frac{1}{2} \leq \frac{3}{2}$$

$$\Rightarrow \left[\cos|x| + \frac{1}{2} \right] = 1 \Rightarrow f(x) = 0 \Rightarrow \int_{x_1}^{x_2} f(x) = 0$$

$$(IV) \text{ Let } AD = x, \angle BAD = \theta$$

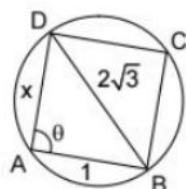
$$\text{In } \triangle ABD, \frac{BD}{\sin \theta} = 2R$$

$$\Rightarrow \sin \theta = \frac{2\sqrt{3}}{4} \Rightarrow \theta = 60^\circ$$

$$\text{using cosine rule } \cos 60^\circ = \frac{x^2 + 1 - BD^2}{2x}$$

$$\Rightarrow BD = \frac{1 + 3\sqrt{5}}{2}$$

$$\therefore a = \frac{1}{2}, b = \frac{3}{2}$$



9. Ans (A)

$$(I) (f(x))^2 + [f'(x)]^2 = h(x)$$

$$h'(x) = 2f(x)f'(x) + h'(x)$$

$$2f'(x)f''(x)$$

$$= 2f'(x)(f(x) + f''(x))$$

$$= 2f'(x)\{-xg(x)f'(x)\}$$

$$= -2x(f'(x))^2g(x)$$

so, $h'(x) < 0$ if $x > 0$ and $h'(x) > 0$ for $x < 0$

Then, $x = 0$ is a maxima

$$(II) f^2(x) + f(x) = g(x)$$

$$2f(x)f'(x) + f(x) = g'(x) \text{ and } g'(x) = f'(x)$$

$$\{2f(x) + 1\}$$

So, $g'(x) \geq 0$ as $g(x)$ is increasing and for

$$f'(x) \geq 0$$

$$2f(x) + 1 \geq 0$$

$$\text{Hence, } f(x) \geq -\frac{1}{2}$$

$$(III) \text{ Let } G(x) = \left[\int_0^x f(t) dt \right]^2 \quad \dots(1)$$

$$G'(x) = 2f(x) \int_0^x f(t) dt$$

$$\text{and } \frac{G'(x)}{2} = g(x)$$

$$\text{and } G'(0) = 0 = g(0)$$

Also, $g(x)$ is non-increasing

$$\text{Hence, } g(x) \geq 0$$

$$x \in (-\infty, 0), g(x) \leq 0 \text{ and } x \in [0, \infty)$$

$$\text{i.e., } G'(x) \geq 0 \text{ for } x \leq 0, G'(x) \leq 0 \text{ for } x \geq 0$$

$$\text{and } G(0) = 0$$

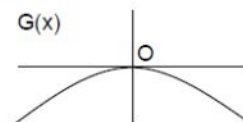
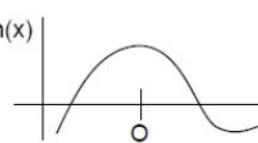
$$\text{But } G(x) \text{ (from equation (1)) } \geq 0$$

$$\text{Hence, } G(x) = 0 \text{ } x \in \mathbb{R}$$

$$\int_0^x f(t) dt = 0 \text{ } x \in \mathbb{R}$$

$$\text{i.e., } f(x) = 0 \text{ } x \in \mathbb{R}$$

$$\text{Hence, } f(1) = 0$$



$$(IV) f(x) = x^2 + \int_0^x e^{-t} f(x-t) dt \quad \dots(1)$$

$$f(x) = x^2 + e^{-x} \int_0^x e^t f(x-x+t) dt$$

$$f(x) = x^2 + e^{-x} \int_0^x e^t f(t) dt$$

$$f'(x) = 2x + e^{-x} [e^x f(x)] - e^{-x} \int_0^x e^t f(t) dt$$

$$f'(x) = 2x + f(x) - \int_0^x e^{-(x-t)} f(t) dt = 2x + f(x) + x^2 - f(x)$$

$$f'(x) = 2x + x^2$$

$$f(x) = x^2 + \frac{x^3}{3} + c \text{ and from equation (1) } f(0) = 0$$

$$\text{Hence, } c = 0$$

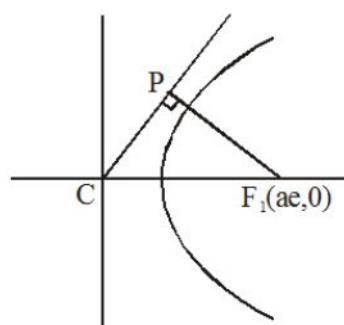
$$\text{i.e., } f(x) = x^2 + \frac{x^3}{3}$$

$$f(-1) = 1 - \frac{1}{2} = \frac{2}{3}, \frac{k}{3} = \frac{2}{3} \therefore k = 2$$

10. **Ans (B)**

Length of perpendicular from $F_1(ae, 0)$ upon

$bx - ay = 0$ is equal to



$$F_1P = \left| \frac{b(ae) - 0}{\sqrt{a^2 + b^2}} \right| = \frac{b\sqrt{a^2 + b^2}}{\sqrt{a^2 + b^2}} = b$$

$\therefore$ Triangle CF_1P is right angle isosceles

$$\therefore CP = F_1P = b \text{ and } CF_1^2 = CP^2 + PF_1^2$$

$$\Rightarrow a^2 e^2 = 2b^2 \Rightarrow a^2 e^2 = 2(a^2 e^2 - a^2)$$

$$\Rightarrow e = \sqrt{2} \Rightarrow a = b$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (1.60)**

$$\frac{xf'(g(x))g'(x)}{f(g(x))} = \frac{g'(f(x))f'(x)}{g(f(x))}$$

$$x \frac{d}{dx} (\ln f(g(x))) = \frac{d}{dx} (\ln g(f(x)))$$

$$\text{Also } \int_0^a f(g(x)) dx = \frac{1}{2} - \frac{e^{-2a}}{2} \quad \forall a$$

$$\therefore \text{ On differentiate, we get } f(g(a)) = e^{-2a} \Rightarrow f(g(x)) = e^{-2x}$$

From Eqns. (1) and (2), we get

$$x \frac{d}{dx} (\ln e^{-2x}) = \frac{d}{dx} (\ln g(f(x)))$$

$$-2x = \frac{d}{dx} (\ln g(f(x)))$$

In integrating, we get

$$\ln g(f(x)) = -x^2 + C$$

$$g(f(x)) = k \cdot e^{-x^2}$$

$$\text{Given } g(f(0)) = 1 \Rightarrow k = 1$$

$$\therefore g(f(4)) = e^{-16} \Rightarrow \lambda = 16$$

$$\frac{\lambda}{10} = 1.60$$

2. **Ans (2.25)**

Centre of circle = $(4, -5)$

$$\text{Radius} = \sqrt{|-4 + 5i|^2 - (-40)} = \sqrt{4^2 + 5^2 + 40} = 9$$

Distance of centre $(4, -5)$ from $(-2, 3)$ is 10.

$$\therefore a = \max. |z - (-2 + 3i)| = 9 + 10 = 19$$

$$b = \min. |z - (-2 + 3i)| = 10 - 9 = 1$$

$$\frac{a-b}{8} = \frac{18}{8} = \frac{9}{4} = 2.25$$

3. Ans (2.25)

$$f(1) + g(1) = 9e; f(x) = -x^2 g'(x) \text{ and } g(x)$$

$$= -x^2 f'(x)$$

$$I = \int_1^4 \frac{f(x) + g(x)}{x^2} dx = \int_1^4 \frac{-x^2 (f'(x) + g'(x))}{x^2} dx$$

$$= \int_1^4 - (f'(x) + g'(x)) dx$$

$$= -[f(x) + g(x)]_1^4 = -(f(4) + g(4) - 9e)$$

$$\text{Now, } f(x) + g(x) = -x^2(g'(x) + f'(x))$$

$$\Rightarrow \frac{g'(x) + f'(x)}{f(x) + g(x)} = \frac{-1}{x^2} \Rightarrow f(1) + g(1) = 9e$$

$$\Rightarrow a = 9$$

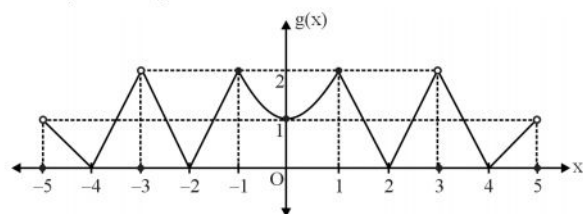
$$\Rightarrow f(x) + g(x) = 9e^{1/x}$$

$$I = -(9e^{1/4} - 9e) = 9(e - e^{1/4}) \equiv k(e - e^{1/4})$$

$$\Rightarrow k = 9$$

$$\frac{k}{4} = \frac{9}{4} = 2.25$$

4. Ans (16.00)



$$g(g(x)) = 1, \text{ let } g(x) = t$$

$$g(t) = 1 \Rightarrow t = 0, t_1, t_2, t_3, t_4, t_5, t_6$$

$$\text{Where } 1 < t_1 < 2, 2 < t_2 < 3, 3 < t_3 < 4, t_4, t_5, t_6 < 0$$

$$\text{Now, } \left. \begin{array}{l} g(x) = 0 \Rightarrow 8 \text{ solutions} \\ g(x) = t_1 \Rightarrow 8 \text{ solutions} \end{array} \right\} \Rightarrow 16 \text{ solutions}$$

5. Ans (14.40)

$$\det.(A^2 + A) = \det.[(P^{-1} DP)(P^{-1} DP) + P^{-1} DP]$$

$$\text{or } \det.(P^{-1} D^2 P + P^{-1} DP)$$

$$\text{or } \det.(P^{-1} (D^2 + D) P)$$

$$\text{or } \det.(P^{-1} P) \cdot \det.(D^2 + D)$$

$$\text{or } \det.(I) \cdot \det.(D) \cdot \det.(D + I)$$

$$\text{or } 1^3 \cdot (1 \cdot 2 \cdot 3) \cdot (2 \cdot 3 \cdot 4) = 6 \cdot 24 = 144$$

6. Ans (9.66 or 9.67)

$$d = \frac{-57 - 303}{30} = -12$$

$$A_r = 303 + r \times (-12) \geq 0 \Rightarrow r \leq \frac{303}{12} \approx 25 \dots$$

$$A_{25} = 303 + 25(-12) = 3$$

$$|A_r|_{\min.} = |A_{25}| = 3$$

$$\frac{S}{(A_{14} - 12)|A_r|_{\min.}} = \frac{29(A_1 + A_{29})}{2(A_{15}) \cdot 3} = \frac{29}{3}$$

7. Ans (30.00)

$$\left(x - 2 + \frac{1}{x}\right)^{30} = n_0 x^{30} + n_1 x^{29} + \dots + n_{29} x + n_{30} + n_{31} x^{-1} + \dots + n_{60} x^{-30}$$

$$(x - 1)^{60} = n_0 x^{60} + n_1 x^{59} + \dots + n_{29} x^{31} + n_{30} x^{30} + n_{31} x^{29} + \dots + n_{60}$$

$$\therefore n_0 = {}^{60}C_0, n_1 = -{}^{60}C_1, n_2 = {}^{60}C_2, \dots,$$

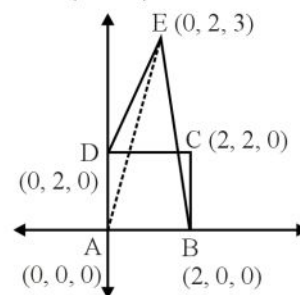
$$n_{30} = {}^{60}C_{30} \dots n_{60} = {}^{60}C_{60}$$

$$\therefore C = n_0 + n_1 + n_2 + \dots + n_{60} = 0$$

$$\text{Hence, } C - n_{30} = -{}^{60}C_{30}$$

$$\therefore a = 30.00$$

8. Ans (2.00)



$$V = \frac{1}{6} [\vec{AB} \quad \vec{AD} \quad \vec{AE}] = 2$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

AIOT

12-05-2024

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY

PAPER-2 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	B	A	C	A				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	B,D	B,D	A,B,C	B,D	A,B,C,D	A,C		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	3	6	5	4	3	3	1	3

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	B	A	B	C				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,B,C,D	A,B,D	D	A,C	A,B,C	A,C,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	9	2	6	4	5	7	7	4

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	B	A	C	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,B,C	A,B,C	A,B,C,D	A,B,D	A,C,D	B,C,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	6	5	6	0	9	4	4	2

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (B)

$$\because r_1 = 2R \sin 30^\circ, r_2 = 2R \sin 60^\circ = R\sqrt{3} = R$$

$$B_D = B_{AC} + B_{CB} = \left(\frac{\mu_0 i}{4\pi r_1} \sin 60^\circ - \frac{\mu_0 i}{4\pi r_2} \sin 30^\circ \right)$$

$$\Rightarrow B_D = \frac{\mu_0 i}{4\pi R\sqrt{3}}$$

4. Ans (A)

$$v = \sqrt{2g(H-h)}$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (B,D)

The observed reading will be = 10 mm + 1 × 01

mm = 1.01 cm

$$\therefore \text{density} = \frac{\text{Mass}}{\text{vol}} = \frac{2.0606}{(1.01)^3} = 1.9999$$

$$\approx 2.00 \text{ gm/cm}^3$$

6. Ans (B,D)

$$F = qE = -q \frac{dv}{dx}$$

$$\frac{1}{2}mv_0^2 = q(10^4)$$

7. **Ans (A,B,C)**

$$E_S = \frac{-GMm}{R}; E_1 = \frac{-GMm}{2(2R)}; E_2 = \frac{-GMm}{2(3R)}$$

$$E_1 - E_S = \frac{3}{4}mgR$$

$$E_2 - E_1 = \frac{mgR}{12}$$

8. **Ans (B,D)**

(B) as temperature increase

(D) as work is done on gas

9. **Ans (A,B,C,D)**

Using conservation of linear momentum and conservation of energy

PART-1 : PHYSICS

SECTION-III

1. **Ans (3)**

If x is the distance of object from the first lens

So, $m_1 = \frac{f_1}{f_1 - x}$ where f_1 is focal length of first lens

and $m_2 = \frac{f_2}{f_2 - \left(L - \frac{xf_1}{f_1 - x}\right)}$ where f_2 is focal

length of the second lens and L is separation between the lenses.

$$\text{So, } \frac{1}{M_1} = \frac{1}{m_1 m_2} = \frac{f_1 - x}{f_1} \cdot \frac{f_2 - \left(L - \frac{xf_1}{f_1 - x}\right)}{f_2}$$

$$= 1 - \frac{L}{f_2} - \frac{x(f_1 + f_2 - L)}{f_1 f_2}$$

When the lenses are interchanged, in the above expression only the second term changes

$$\text{So, } \frac{1}{M_2} = 1 - \frac{L}{f_1} - \frac{x(f_1 + f_2 - L)}{f_1 f_2}$$

$$\therefore \frac{1}{M_1} - \frac{1}{M_2} = L \left(\frac{1}{f_1} - \frac{1}{f_2} \right) = L(D_1 - D_2)$$

$$\text{So, } 1 - \frac{1}{4} = \frac{1}{4}(D_1 - D_2)$$

$$\therefore D_1 - D_2 = 3 \text{ diopter}$$

2. **Ans (6)**

$$r_{\text{cell}} = \frac{24}{24} = 1\Omega$$

$$\frac{24}{1.6} = 15 = 2x + R + 1$$

$$\frac{24}{2} = 12 = 2x + 1 + \frac{R(14 - 2x)}{R + (14 - 2x)}$$

solving $R = 6\Omega$

3. **Ans (5)**

$$f = \frac{1}{2l} \sqrt{\frac{T}{\rho A}} \dots\dots(1)$$

$$\frac{m}{Al} = \rho \dots\dots(2)$$

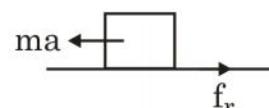
$$\sigma = \frac{T}{A} \dots\dots(3)$$

$$\therefore f = \frac{T}{2m} \sqrt{\frac{\rho}{\sigma}} = 375 \text{ Hz}$$

4. **Ans (4)**

$$V = 2t^2$$

$$a = 4t$$



at $t = 1$ sec slipping occurs

$$ma = \mu_s mg$$

$$4t = \mu_s g$$

$$\mu_s = 0.4$$

$t = 1 \rightarrow t = 3$ slipping occurs

b/w $t = 1$ and $t = 2$ sec.

$$(4t - \mu_k g) = \frac{dv}{dt}$$

$$v = [2t^2 - \mu_k g t]_1^2 = 2 \times 3 - \mu_k g$$

$$\mu_k = 0.3 \frac{3\mu_s}{\mu_k} = 4$$

5. **Ans (3)**

$$\epsilon_{\text{im}} = \frac{1}{3} \pi a^2 N \omega B_0 = 3$$

6. **Ans (3)**

For ring just slides on to the steel rod the diameter of rod and ring should be equal to each other and suppose due to AG increment in temperature the diameter of both are equal then

$$4(1 + \alpha_S \Delta\theta) = 3.992(1 + \alpha_{\text{Brass}} \Delta\theta)$$

$$4 + 4 \times 11 \times 10^{-6} \times \Delta\theta = 3.992 + 3.992 \times 20 \times 10^{-6} \times \Delta\theta$$

$$4 + 44 \times 10^{-6} \Delta\theta = 3.992 + 79.84 \times 10^{-6} \times \Delta\theta$$

$$0.008 = 35.84 \times 10^{-6} \Delta\theta$$

$$\frac{8 \times 10^3}{35.84} = \Delta\theta; \Delta\theta = \frac{8000}{35.84} = 283$$

so if temperature increased by 223°C then ring will start to slide and this temperature will equal to

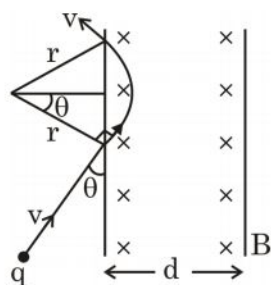
$$\theta = 30^\circ + \Delta\theta = 30 + 253 = 283^\circ\text{C}$$

$$\theta = 283^\circ\text{C} \approx 280^\circ\text{C}$$

7. **Ans (1)**

For particle to come back to left side,

$$d > (r - r \cos\theta)$$



$$d > \frac{mv}{qB}(1 - \cos\theta)$$

$$m = 1 \text{ mg} = 10^{-6} \text{ kg}$$

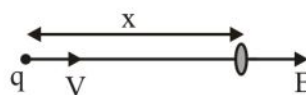
$$v = 50 \text{ m/s}, q = 10^{-6} \text{ C}, B = 10 \text{ T}$$

$$\cos\theta = 4/5$$

$$\Rightarrow d > \frac{10^{-6} \times 50}{10^{-6} \times 10} \left(1 - \frac{4}{5}\right)$$

$$\Rightarrow d > 1 \text{ m}$$

8. **Ans (3)**



$$\phi_E = E \cdot A = \left(\frac{1}{4\pi\epsilon_0}\right) \frac{q}{x^2} \cdot A$$

$$I_d = \left| \epsilon_0 \cdot \frac{d\phi_E}{dt} \right|$$

$$= \left| \epsilon_0 \cdot \frac{qA}{4\pi\epsilon_0} \frac{d}{dt} \left(\frac{1}{x^2}\right) \right| = \frac{qAv}{2\pi x^3}$$

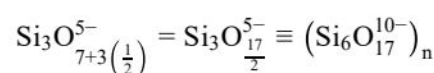
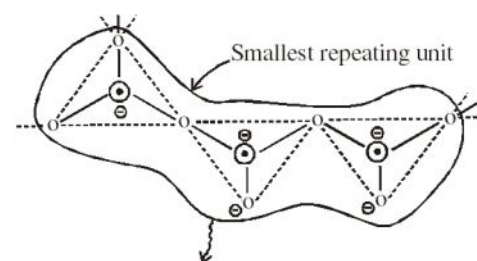
PART-2 : CHEMISTRY

SECTION-I (i)

2. **Ans (A)**

$$\frac{\ell \ln 2}{K} = \frac{1}{K} \ell \ln \left[\frac{r_\infty - r_0}{r_\infty - r_t} \right]$$

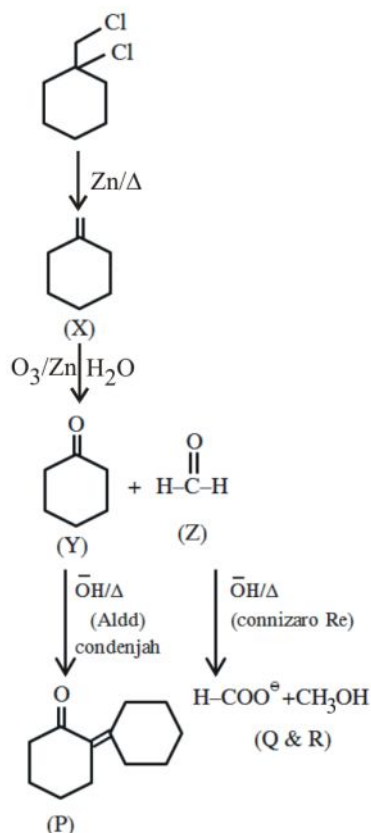
4. **Ans (C)**



PART-2 : CHEMISTRY

SECTION-I (ii)

5. Ans (A,B,C,D)



7. Ans (D)

Below CMC, the solution is true solution.

8. Ans (A,C)

Net cell reaction is



$$\Delta G_{50^\circ\text{C}}^\circ = -nFE^\circ = -2 \times 96500 \times 0.6 = -115800 \text{ J}$$

$$\Delta G_{0^\circ\text{C}}^\circ = -nFE^\circ = -2 \times 96500 \times 0.7 = -135100 \text{ J}$$

$$\Delta S^\circ = nF \cdot \left(\frac{E_2^\circ - E_1^\circ}{T_2 - T_1} \right) = 2 \times 96500 \times \frac{0.6 - 0.7}{50} = -386 \text{ J/K}$$

$$\Delta H^\circ = \Delta G^\circ + T \cdot \Delta S^\circ$$

$$= (-135100) + 273 \times (-386) = 240478 \text{ J}$$

9. Ans (A,B,C)

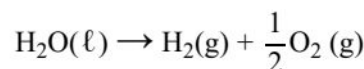
Oxidising strength = $\text{Ti}^{+3} < \text{Fe}^{+3} < \text{Mn}^{+3}$

PART-2 : CHEMISTRY

SECTION-III

1. Ans (9)

$$P_{\text{N}_2} = 1 - 0.04 = 0.96 \text{ atm}$$



Initial	x mole	0	0
Final	0	x	x/2

Now,

$$\frac{3x}{2} = \frac{(1.9 - 0.96 - 0.04) \times 20}{0.08 \times 300}$$

$$\Rightarrow x = 0.5$$

$\therefore$ Mass of liquid water present initially
 $= x \times 18 = 9 \text{ gm}$

2. Ans (2)

$$[\text{Cd}^{2+}] = \frac{0.1 \times 10^{-3}}{10} \times 1000 = 10^{-2} \text{ M}$$

$$[\text{S}^{2-}] = \frac{10^{-28}}{10^{-2}} = 10^{-26}$$

$$10^{-29} = \frac{10^{-26} [\text{H}^+]^2}{0.1}$$

$$[\text{H}^+] = 10^{-2}$$

3. Ans (6)

$$2\pi r = 3\pi \Rightarrow \frac{2\pi r}{\lambda} = 3 = n$$

T.E. in the orbit = -13.6 eV

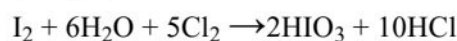
$$\Rightarrow -\frac{13.6 Z^2}{3^2} = -13.6 \Rightarrow Z = 3, \text{ Sample is } \text{Li}^{2+}.$$

Ground state energy = -13.6 $\times$ 9 eV

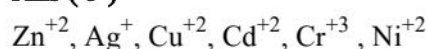
Energy of ejected e^- = 126.566 - 122.4 = 4.166

$$\lambda = \left(\frac{150}{4.16} \right)^{\frac{1}{2}} = 6 \text{ \AA}$$

4. Ans (4)



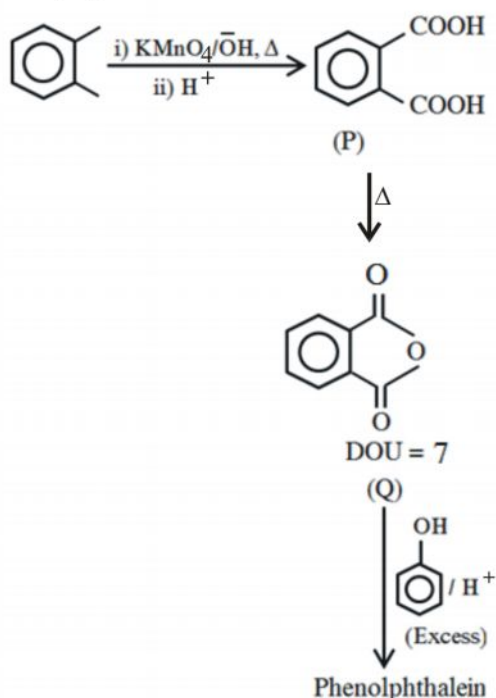
5. Ans (5)



6. Ans (7)

i, ii, iii, iv, v, vi, vii

7. Ans (7)



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B)

$$\log \pi (\sin^{-1}(x) (\cos^{-1}(x))) = \log \left(\frac{\pi^2}{18} \right)$$

$$\Rightarrow \sin^{-1}(x) \left(\frac{\pi}{2} - \sin^{-1}(x) \right) = \frac{\pi^2}{18}$$

$$\Rightarrow (\sin^{-1}(x))^2 - \frac{\pi}{2} (\sin^{-1}(x)) + \frac{\pi^2}{18} = 0$$

$$\Rightarrow \left(\sin^{-1}(x) - \frac{\pi}{3} \right) \left(\sin^{-1}(x) - \frac{\pi}{6} \right) = 0$$

$$\Rightarrow \sin^{-1}(x) = \frac{\pi}{3} \text{ or } \frac{\pi}{6}$$

$$x = \frac{\sqrt{3}}{2} \text{ or } \frac{1}{2}$$

$$x_1^2 + x_2^2 = \frac{3}{4} + \frac{1}{4} = 1$$

$$\text{So } \tan^{-1}(1) = \frac{\pi}{4}$$

2. Ans (A)

$$(3 + i)^n = (a_n + b_n i)$$

squaring both side

$$(8 + 6i)^n = a_n^2 - b_n^2 + 2a_n b_n i$$

$$\Rightarrow \left(\frac{8 + 6i}{12} \right)^n = \frac{a_n^2 - b_n^2}{12^n} + \frac{2a_n b_n i}{12^n}$$

$$\Rightarrow \sum_{n=0}^{\infty} \frac{a_n b_n}{12^n} = \frac{1}{2} \sum_{n=0}^{\infty} \text{Im} \left(\frac{8 + 6i}{12} \right)^n$$

$$= \frac{1}{2} \left[\frac{1}{1 - \left(\frac{4+3i}{6} \right)} \right]$$

$$= \frac{1}{2} \left[\frac{6}{2-3i} \right] = \frac{1}{2} 6 \left[\frac{2+3i}{13} \right] = \frac{9}{13}$$

3. Ans (C)

$$\Rightarrow \int_0^{\frac{\pi}{4}} \frac{\tan x - [\tan x]}{\tan x} dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\tan x - [\tan x]}{\tan x} dx$$

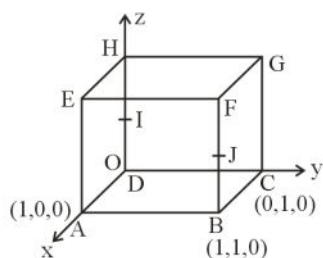
$$\Rightarrow \int_0^{\frac{\pi}{4}} 1 dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} 1 - (\cot x) dx$$

$$= \frac{\pi}{4} + \left(\frac{\pi}{3} - \frac{\pi}{4} \right) - (\ln \sin x)_{\pi/4}^{\pi/3}$$

$$= \frac{\pi}{3} - \left(\ln \frac{\sqrt{3}}{2} - \ln \frac{1}{\sqrt{2}} \right)$$

$$= \frac{\pi}{3} - \frac{1}{2} \ln \left(\frac{3}{2} \right)$$

4. Ans (B)



$$E(1, 0, 1), I\left(0, 0, \frac{1}{2}\right),$$

$$J\left(1, 1, \frac{1}{2}\right), C(0, 1, 0)$$

$$\Delta EIJ + \Delta IJC$$

$$\vec{EI} = \left(1, 0, \frac{1}{2}\right) \quad \vec{EJ} = \left(0, -1, \frac{1}{2}\right)$$

$$\Delta = \frac{1}{2} \left| \vec{EI} \times \vec{EJ} \right|$$

$$= \frac{1}{2} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & \frac{1}{2} \\ 0 & -1 & \frac{1}{2} \end{vmatrix}$$

$$= \frac{1}{2} \left| \hat{i} \left(\frac{1}{2} \right) - \hat{j} \left(\frac{1}{2} \right) + \hat{k} (-1) \right|$$

$$= \frac{1}{2} \sqrt{\frac{1}{4} + \frac{1}{4} + 1} = \frac{1}{2} \sqrt{\frac{3}{2}}$$

$$\Rightarrow \text{Total surface area} = \frac{\sqrt{3}}{2}$$

$$\Rightarrow R = \sqrt{\frac{3}{2}} \quad R^2 = \frac{3}{2}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,B,C)

$$\sum_{i=3}^{40} \binom{i}{2} = \sum_{i=3}^{40} \left(\frac{i(i-1)}{2} \right)$$

$$= \sum_{i=3}^{40} \frac{i(i-1)}{2} \left(\frac{i(i-1)}{2} - 1 \right)$$

$$= \frac{1}{8} \sum_{i=3}^{40} i(i-1)(i(i-1)-2)$$

$$= \frac{1}{8} \sum_{i=3}^{40} i(i-1)(i^2-i-2)$$

$$= \frac{1}{8} \sum_{i=3}^{40} i(i-1)(i+1)(i-2)$$

$$= \frac{1}{8} \sum_{i=3}^{40} i(i-2)(i-1)i(i+1)$$

$$= \frac{1}{40} \sum [(i-2)(i-1)i(i+1)(i+2) - (i-3)(i-2)(i-1)i(i+1)]$$

$$= \frac{38.39.40.41.42 - 0}{40}$$

$$= 38 \cdot 39 \cdot 41 \cdot 42$$

$$= 19 \times 2^2 \times 13 \times 3 \times 41 \times 7 \times 3$$

$$S = 2^2 \times 3^2 \times 7 \times 13 \times 19 \times 41$$

6. Ans (A,B,C)

$$12|z|^2 = 2|z+2|^2 + |z^2+1|^2 + 31$$

$$\Rightarrow 12z\bar{z} = 2(z+2)(\bar{z}+2) + (z^2+1)(\bar{z}^2+1) + 31$$

$$\Rightarrow 12z\bar{z} = 2(z\bar{z} + 2(z+\bar{z}) + 4) + z^2\bar{z}^2 + z^2 + \bar{z}^2 + 1 + 31$$

$$\Rightarrow [(z^2 + 2z\bar{z} + \bar{z}^2) + 4(z+\bar{z}) + 4] + z^2\bar{z}^2 - 12z\bar{z} + 36 = 0$$

$$\Rightarrow (z + \bar{z} + 2)^2 + (z\bar{z} - 6)^2 = 0$$

$$\Rightarrow z + \frac{6}{z} = z + \bar{z} = -2$$

7. Ans (A,B,C,D)

Clearly, EF is parallel to BC.

Since all the faces of ABCD are equilateral triangles,

we have

$$AE = AF = EF = 4, AD = AB = AE + BE = 7$$

By the law of cosines,

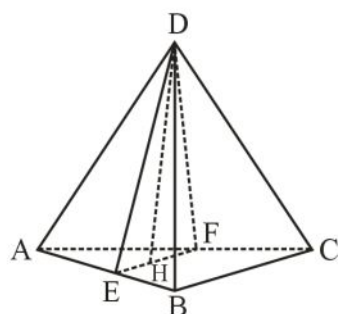
$$DE = \sqrt{AD^2 + AE^2 - 2AD \cdot AE \cdot \cos 60^\circ}$$

$$= \sqrt{49 + 16 - 28} = \sqrt{37},$$

and $DF = \sqrt{37}$ likewise.

Let DH be an altitude of isosceles $\triangle DEF$,

H being the foot on EF, as shown in the figure

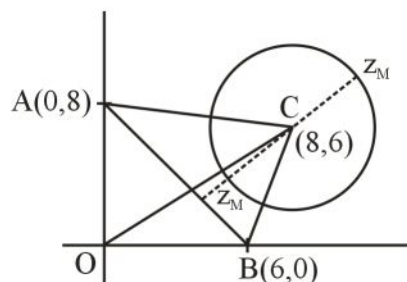


Then $EH = \frac{1}{2}EF = 2$, and

$$DH = \sqrt{DE^2 - EH^2} = \sqrt{33}$$

We conclude $S_{\triangle DEF} = \frac{1}{2} \cdot EF \cdot DH = 2\sqrt{33}$.

8. Ans (A,B,D)



$$\text{Let } x - 8 = 3\cos\theta, y - 6 = 3\sin\theta$$

$$\text{Area} = \frac{1}{2} \begin{vmatrix} 0 & 8 & 1 \\ 6 & 0 & 1 \\ 3\cos\theta + 8 & 3\sin\theta + 6 & 1 \end{vmatrix} \Delta = \frac{1}{2}$$

$$[18\sin\theta + 36 + 24\cos\theta + 52]$$

$$\Delta_{\max} = \frac{1}{2} [52 + \sqrt{18^2 + 24^2}]$$

$$= 41$$

$$\Delta_{\max} = 41, \Delta_{\min} = 11$$

$$\text{for } z_M + \bar{z}_M = 2\text{Re}(z_M) = \frac{104}{5}$$

$$z_M = \left[\frac{52}{5}, \frac{39}{5} \right]$$

$$\text{for } z_M + \bar{z}_M = 2\text{Im}(z_M) = \frac{78}{5}$$

and slope of OC $< \frac{\pi}{4}$

9. Ans (A,C,D)

Put $x = n$ and $y = 1$ then,

$$f(n+1) = \left(1 + \frac{1}{n+1}\right) f(n) + \left(1 + \frac{n}{2}\right) \frac{3}{2} + n^2 + 2n$$

Upon rearranging, one gets

$$\frac{f(n+1)}{n+2} - \frac{f(n)}{n+1} = n + \frac{3}{4}$$

Then we have

$$\frac{f(n)}{n+1} - \frac{f(n-1)}{n} = n - 1 + \frac{3}{4},$$

$$\frac{f(n-1)}{n} - \frac{f(n-2)}{n-1} = n - 2 + \frac{3}{4},$$

∴

$$\frac{f(2)}{3} - \frac{f(1)}{2} = 1 + \frac{3}{4}$$

Adding these equalities together, we get

$$\begin{aligned} \frac{f(n)}{n+1} - \frac{f(1)}{2} &= 1 + 2 + \dots + (n-1) + \frac{3}{4}(n-1) \\ &= \frac{(n-1)n}{2} + \frac{3}{4}(n-1) \end{aligned}$$

Thus,

$$\begin{aligned} f(n) &= (n+1) \left[\frac{(n-1)n}{2} + \frac{3}{4}(n-1) + \frac{1}{2} \cdot \frac{3}{2} \right] \\ &= \frac{n(n+1)(2n+1)}{4} \end{aligned}$$

Now check the options

10. Ans (B,C,D)

$$y' \sin x + y \cos x = x \sin x$$

$$\Rightarrow y' + y \cot x = x \Rightarrow y \sin x = \int x \sin x \, dx + c$$

$$\Rightarrow y \sin x = x \cos x + \sin x + c$$

$$\Rightarrow y = 1 - x \cot x \quad \left(\text{as } y \left(\frac{\pi}{2} \right) = 1 \right)$$

$$\text{Now, } y'(x) = x \operatorname{cosec}^2 x - \cot x$$

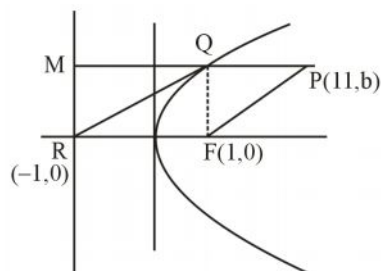
$$= \frac{x - \sin x \cos x}{\sin^2 x} = \frac{2x - \sin 2x}{2 \sin^2 x} > 0 \quad \forall x \in \left(0, \frac{\pi}{2}\right)$$

Hence $y(x)$ is strictly increasing in $\left(0, \frac{\pi}{2}\right)$

PART-3 : MATHEMATICS

SECTION-III

1. Ans (6)



$$(FQ + QP)_{\min} = (MQ + QP)_{\min}$$

M, Q, P must be collinear

$$M(-1, 2t), Q(t^2, 2t), P(11, b)$$

$$\Delta = \begin{vmatrix} -1 & 2t & 1 \\ t^2 & 2t & 1 \\ 1 & b & 1 \end{vmatrix} = 0$$

$$\text{or } \boxed{b = 2t}$$

If FQPR is a parallelogram then slope of FP

= slope of QR

$$\frac{b-0}{10} = \frac{2t}{t^2+1} \Rightarrow \boxed{t=3}$$

$$\text{Area} = 2 \times 3 = 6$$

2. Ans (5)

$$\det(A + xB) = \frac{1}{2} [(Tr(A + xB))^2 - Tr((A + xB)^2)]$$

$$= \frac{1}{2} [(Tr(A) + xTr(B))^2 - Tr(A^2 + xAB + xBA + B^2x^2)]$$

$$= \frac{1}{2} [(Tr(A))^2 + 2Tr(A) Tr(B)x + (Tr(B))^2x^2 - Tr(A^2) - 2Tr(AB)x - Tr(B^2)x^2]$$

$$= \frac{1}{2} [(Tr(A))^2 - Tr(A^2)] + (Tr(A) Tr(B) - Tr(AB))x + \frac{1}{2} [(Tr(B))^2 - Tr(B^2)]x^2$$

$$= \det A + (Tr(A) Tr(B) - Tr(AB))x + (\det B)x^2,$$

$$272 = 22 + (28 - 33)x + 11x^2$$

$$\Rightarrow 11x^2 - 5x - 250 = 0$$

$$\Rightarrow x = -\frac{50}{11} \text{ or } 5$$

3. Ans (6)

Let α be an integer satisfying the given equation, so that

$$4\alpha^2 - (4\sqrt{3} + 4)\alpha + \sqrt{3}n - 24 = 0$$

$$\Rightarrow 4\alpha^2 - 4\alpha - 24 = \sqrt{3}(4\alpha - n).$$

Since $\sqrt{3}$ is irrational, it follows that

$$4\alpha^2 - 4\alpha - 24 = 0 \text{ and } 4\alpha - n = 0$$

Substituting $\alpha = \frac{n}{4}$ into the first equation, we have

$$4\left(\frac{n}{4}\right)^2 - 4\left(\frac{n}{4}\right) - 24 = 0$$

$$\Rightarrow n^2 - 4n - 96 = 0$$

$$\Rightarrow (n - 12)(n + 8) = 0$$

$$\Rightarrow n = 12, -8$$

Since n is a positive integer, we have $n = 12$

4. Ans (0)

$$I = \int_0^{\infty} \frac{1}{(1 + x^{2024})(1 + x^2)} dx \quad \dots(1)$$

$$\text{Put } x = \frac{1}{t}$$

$$I = \int_{\infty}^0 \frac{x^{2024} \cdot x^2}{(1 + x^{2024})(1 + x^2)} \left(-\frac{1}{x^2}\right) dx$$

$$I = \int_0^{\infty} \frac{x^{2024}}{(1 + x^{2024})(1 + x^2)} dx \quad \dots(2)$$

$$2I = \int_0^{\infty} \frac{1}{(1 + x^2)} dx = \frac{\pi}{2}$$

$$I = \frac{\pi}{4} \text{ or } \left[\frac{\pi}{4}\right] = 0$$

5. Ans (9)

There are $5 \cdot 6^3$ functions for which $f(1) \neq a$ and another $5 \cdot 6^3$ functions for which $f(2) \neq b$. There are

$5^2 \cdot 6^2$ functions for which both $f(1) \neq a$ and $f(2) \neq b$.

So the total number of functions for which $f(1) \neq a$

or $f(2) \neq b$ or both is $5 \cdot 6^3 + 5 \cdot 6^3 - 5^2 \cdot 6^2 = 1260$

6. Ans (4)

Let A denote the event that the purchase accepts

a lot. Now,

$$P(A) = P(A|\text{lot has } x \text{ defectives}) \frac{3}{10} + P(A|\text{lot}$$

$$\text{has 1 defective}) \frac{7}{10}$$

$$\frac{54}{100} = \frac{{}^x C_0 ({}^{10-x} C_3)}{{}^{10} C_3} \left(\frac{3}{10}\right) + \frac{{}^1 C_0 ({}^9 C_3)}{{}^{10} C_3} \left(\frac{7}{10}\right)$$

$$\Rightarrow \frac{54}{100} \times {}^{10} C_3 = \frac{(10-x)(9-x)(8-x)}{6} \left(\frac{3}{10}\right) + \frac{9 \times 8 \times 7}{6} \times \frac{7}{10}$$

$$\Rightarrow \frac{54}{100} \times \frac{10 \times 9 \times 8}{6} \times 60 = (10-x)(9-x)$$

$$(8-x)(3) + 9 \times 8 \times 7 \times 7$$

$$\Rightarrow (10-x)(9-x)(8-x) = 120$$

$$\Rightarrow x = 4$$

7. Ans (4)

$$T_n + 2 = 2\sqrt{2S_n}$$

$$\Rightarrow T_n^2 + 4T_n + 4 = 8S_n \quad \dots(1)$$

$$\Rightarrow T_{n-1}^2 + 4T_{n-1} + 4 = 8S_{n-1} \quad \dots(2)$$

by (1) & (2)

$$\Rightarrow T_n^2 - T_{n-1}^2 + 4(T_n - T_{n-1}) = 8(S_n - S_{n-1})$$

$$\Rightarrow T_n^2 - T_{n-1}^2 + 4(T_n - T_{n-1}) = 8T_n$$

$$\Rightarrow (T_n + T_{n-1})(T_n - T_{n-1}) = 4(T_n + T_{n-1})$$

$$\Rightarrow T_n - T_{n-1} = 4$$

so sequence is A.P. with common difference 4

by equation (1) put $n = 1$, $S_1 = T_1$

$$T_1 = 2, d = 4$$

$$\therefore T_{2024} = 2 + (2023)4 = 8094$$

8. Ans (2)

Put $x = ty$

$$g(t) = \int_{-\frac{1}{t}}^{\frac{1}{t}} f(y) t dy = t \int_{-\frac{1}{t}}^{\frac{1}{t}} f(y) dy$$

$$\frac{g(t)}{t} = 2 \int_0^{\frac{1}{t}} f(y) dy$$

$$= 2 \int_0^{\frac{1}{t}} e^{-y} dy$$

$$= -2[e^{-y}]_0^{1/t} \Rightarrow -2[e^{-1/t} - 1]$$

$$\lim_{t \rightarrow 0^+} \frac{g(t)}{t} \Rightarrow 2$$